ClipSAT : Qiyas English GAT Quantitative & SAAT Mathematics

Unit 8.01: SAAT Limits, Derivatives, Definite Integrals & Area Name: _____

Q1. The coordinate diagram shows the parabolic curve f_1x = x^2 - 20x + 118 in the domain of SAAT Differential & Integral Calculus. By completing the square f_1x = x - h^2 + k , find the sum k + f_112 .

[ ] A: 50
[ ] B: 22
[ ] C: 38
[ ] D: 40

Q2. A mathematical model in SAAT Differential & Integral Calculus is defined by the cubic function y_2x = 3x^3 - 10x^2 + 15x - 15 . Calculate the exact gradient of the tangent to the curve at the point where x = 3 .

[ ] A: 36
[ ] B: 96
[ ] C: 21
[ ] D: 22

Q3. In SAAT Differential & Integral Calculus, two displacement vectors are given by mathbf{u}_{3} = langle 10, -8 rangle and mathbf{v}_{3} = langle 9, 7 rangle . Compute the exact scalar value of mathbf{u}_{3} cdot mathbf{v}_{3} + |mathbf{u}_{3}|^2 .

[ ] A: 34
[ ] B: 198
[ ] C: 164
[ ] D: 310

Q4. The linear growth profile shown in the diagram for SAAT Differential & Integral Calculus follows an arithmetic progression with first term a_1 = 11 and common difference d = 6 . Find the exact sum S_{15} of the first 15 terms.

[ ] A: 95
[ ] B: 885
[ ] C: 795
[ ] D: 1425

Q5. In SAAT Differential & Integral Calculus, evaluate the exact definite integral I_5 = int_0^{2} 18x^2 + 40x,mathrm{d}x .

[ ] A: 152
[ ] B: 168
[ ] C: 65
[ ] D: 128

Q6. In SAAT Differential & Integral Calculus, consider the quadratic function f_6x = x^2 - 22x + 211 . Express f_6x in the completed-square form x - h^2 + k and compute the exact value of k + f_618 .

[ ] A: 229
[ ] B: 240
[ ] C: 139
[ ] D: 222

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Q7. The graph illustrates the cubic model y_7x = 3x^3 - 11x^2 + 33x - 22 studied in SAAT Differential & Integral Calculus. Find the instantaneous gradient frac{mathrm{d}y}{mathrm{d}x} of the tangent to the curve at x = 8 .

[ ] A: 609
[ ] B: 433
[ ] C: 400
[ ] D: 83

Q8. In SAAT Differential & Integral Calculus, two displacement vectors are given by mathbf{u}_{8} = langle 11, -7 rangle and mathbf{v}_{8} = langle 8, 5 rangle . Compute the exact scalar value of mathbf{u}_{8} cdot mathbf{v}_{8} + |mathbf{u}_{8}|^2 .

[ ] A: 53
[ ] B: 170
[ ] C: 223
[ ] D: 293